

Article

Large Language Models for Coagulant Dosage Prediction: A Systematic Investigation of Generalization, Rationale Patterns and Hallucination

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Abstract

Precise coagulant dosage control is essential for stable drinking-water treatment, yet conventional machine learning (ML) methods can be sensitive to data conditions. This study evaluates a large language model (LLM)-based in-context workflow for this tabular prediction task using the DeepSeek family, benchmarked against XGBoost, ANN, SVM, and k-NN under a shared chronological protocol. We examined performance across feature configurations, training-pool conditions, and outlier subsets. On the HQ-WTP case dataset, full-feature input outperformed temperature-only input, indicating the value of multivariate information. Performance responses to training-pool condition were model-dependent, with no universal optimum. Under the fixed protocol, in the full-feature test setting, the strongest tabular baseline showed the strongest test performance, while DeepSeek-Reasoner and DeepSeek-Chat showed intermediate performance, and DeepSeek-R1-Distill-Qwen-32B showed relatively lower stability. DeepSeek-Reasoner reached its best test performance at the 1/2 condition. We also coded narrative rationale patterns in generated responses and performed a protocol-based hallucination audit. DeepSeek-Reasoner showed comparatively lower hallucination incidence and more stable error behavior in this benchmark. These analyses are interpreted as response-level reliability evidence rather than verification of internal computational mechanisms. Overall, the study provides transparent, case-specific benchmark evidence for LLM-assisted decision support, while broader deployment claims require strict hallucination control, human-in-the-loop safeguards, and independent multi-site validation.

Keywords: coagulant dosage prediction; large language models (LLMs); tabular regression benchmark; hallucination analysis; output-level rationale coding; robustness evaluation



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1. Introduction

Coagulant dosage serves as a critical step in drinking-water treatment. Its efficacy directly determines the operational performance of subsequent sedimentation and filtration processes, as well as the final effluent water quality [1,2]. By dosing coagulants such as aluminum or iron salts, the coagulation process neutralizes the surface charge of colloidal particles. This promotes the aggregation of micro-flocs into settleable flocs, thereby efficiently removing pollutants such as turbidity and organic matter [3,4].

Historically, dosage decisions relied heavily on operator experience and sensory judgments of water quality (e.g., color, turbidity) [2,5,6]. This manual approach is inherently

subjective and suffers from significant lags and an inability to cope with sudden water quality fluctuations, leading to chemical wastage [2,4] and unstable effluent quality [5,7,8]. While jar tests can enhance accuracy, they are labor-intensive and time-consuming, making it difficult to respond to emergency water quality variations [9]. Furthermore, some water-treatment plants may face challenges due to insufficient testing frequency [10].

Traditional automation methods also face limitations. Feedforward control struggles with the nonlinear dynamics of raw-water parameters, while traditional PID feedback control is often compromised by hydraulic retention times (process time lags), thus failing to achieve stable control targets [11,12].

Artificial intelligence (AI) is increasingly applied in civil and environmental engineering [13–15]. For drinking-water coagulation specifically, AI has begun to reshape coagulant dosing strategies [16]. Generally, these approaches can be categorized into prediction models and decision-optimization models [5]. Predictive approaches utilizing Artificial Neural Networks (ANNs), Extreme Learning Machines (ELM), and fuzzy logic have significantly reduced prediction errors by mapping nonlinear relationships between water quality parameters [5,6,17]. Optimization algorithms, such as Genetic Algorithms (GAs), can generate multi-objective Pareto optimal solution sets to maximize operational benefits [18]. Furthermore, several studies have employed evolutionary algorithms to optimize the hyperparameters of prediction models [7,19]. For instance, Wang et al. [7] proposed a control model that used the Improved Shuffled Frog Leaping Algorithm (ISFLA) to optimize the parameters of Support Vector Machines (SVMs). Similarly, Zhang et al. [19] established a hybrid HANN using water quality and operational parameters as inputs, in which weights and biases of the network were optimized by the GA. As a result, the model accuracy was successfully improved.

Recently, large language models (LLMs) have demonstrated transformative potential across diverse fields, including software engineering [20,21], healthcare [22,23], transportation and urban planning [24], chemistry and material science [25–27], energy and environmental protection [28,29], and social behavior analysis [30,31]. In the water sector, the applications of LLMs have been increasing, covering areas such as water quality monitoring, anomaly diagnosis, and energy optimization [32–34]. These studies demonstrate the potential of LLMs to enhance the intelligence of water systems, reduce the burden of manual coding, and optimize resource utilization efficiency.

However, this study addresses a different problem setting from the text-centric tasks where LLMs are commonly evaluated. Coagulant dosage prediction here is a tabular regression task that maps structured physicochemical and operational variables to a continuous target under nonlinear and lag-related process constraints. Accordingly, we treat LLMs as prompt-conditioned predictors on structured records and do not assume that strengths in text tasks transfer directly to tabular control prediction.

In this context, the use of LLMs for coagulant dosage decision support remains insufficiently characterized. To address this gap, we develop an in-context prediction workflow and compare it with conventional tabular ML baselines under the same evaluation protocol. We further examine performance sensitivity to feature configuration, training-pool condition, and outlier subsets, and audit generated rationale text and hallucination patterns as response-level phenomena rather than verified internal mechanisms. This study provides case-specific benchmark evidence for natural-language-mediated decision support in water treatment.

2. Methods

2.1. Study Area and Data Acquisition System Overview

This study was conducted at the HQ water-treatment plant (WTP). The core infrastructure for the coagulation and sedimentation process was a folded-plate flocculation and horizontal-flow sedimentation tank. The process comprised three main stages: mechanical mixing, folded-plate flocculation, and horizontal sedimentation. Liquid Polyaluminum Chloride (PAC) was selected as the coagulant, dosed via an automated control system.

Eight key indicators were monitored: influent turbidity (NTU), influent pH, influent temperature ($^{\circ}\text{C}$), influent flow rate (m^3/h), influent conductivity (S/m), influent dissolved oxygen (mg/L), influent ammonia nitrogen (mg/L), and settled turbidity. The control target was settled turbidity. The coagulant dosage, measured in kg/kt , served as the core prediction parameter for the dosing system.

The coagulant dosage is defined as the mass of commercial PAC applied per unit volume of treated water. It is calculated from the dosing-pump volumetric flow rate together with the flocculant concentration mix ratio and the stock-solution density, which yields the PAC mass flow rate; this mass flow rate is then divided by the total flocculation influent flow rate on the same time basis to obtain the dosage.

PAC dose and settled turbidity were not treated as synchronous at the same clock hour. The transport delay from the dosing point to the settled-turbidity control point was estimated from a discrete hourly water balance on the effective volume (V_0) of the flocculation–sedimentation train. Let Q_i denote the influent flow rate in hour i ($\text{m}^3 \cdot \text{h}^{-1}$), indexed forward from the dosing event. After each full hour $i = 1, \dots, k$, the remaining volume was updated as

$$V_i = V_{i-1} - Q_i \quad (1)$$

with V_0 equal to the total effective volume. The integer k was taken as the number of full hours elapsed until V_k/Q_{k+1} first fell to not more than 1. The continuous travel time from the dose to the settled-turbidity location was then

$$t_r = k + V_k/Q_{k+1} \text{ (h)} \quad (2)$$

All supervisory variables, including PAC dose, were recorded on hourly timestamps. The continuous lag t_r was therefore converted to an integer hour offset $\Delta t = \text{round}(t_r)$, i.e., rounded to the nearest whole hour, and each hourly modeling record paired the settled-turbidity (control) signal at time t with the PAC dose at time $t - \Delta t$ on the same hourly grid.

Data were obtained from continuous monitoring records at HQ WTP. The sampling frequency was one-hour intervals, and the initial dataset contained 1200 records. During data cleaning, observations attributable to equipment malfunction were removed (including rows with all-zero readings and values that were abnormally large or abnormally small), leaving 995 valid samples for analysis.

This study relies on supervisory records from a single drinking-water-treatment plant (HQ WTP), with 995 quality-controlled hourly samples after fault-related filtering. Findings are therefore presented as a single-site case study; extending models or recommendations to other plants, raw-water regimes, seasons, or operating policies would require independent datasets, longer horizons, and purpose-built validation protocols beyond those reported here.

2.2. Construction of an LLM-Based Framework for Coagulant Dosage Prediction via In-Context Learning

To systematically evaluate the performance and generalization capability of LLMs in coagulant dosage prediction, this study designed the prediction framework illustrated

in Figure 1. We adopted an in-context learning (ICL) strategy, where historical data with known dosage labels served as demonstration examples. This allowed the model to infer underlying patterns within the data without updating its internal parameters [35]. Because this paradigm differs from conventional parameter-fitting training, the train/test split is used here as an evaluation protocol for held-out prediction behavior, not as evidence of method-equivalent learning mechanisms. The training set, containing dosage labels, constructed the prompt context to guide the model in capturing the complex mapping between input variables and dosage. Conversely, the test set lacked dosage labels and was reserved exclusively for evaluating prediction performance on unseen samples. To avoid random splitting, potentially leaking temporal structure and overstating performance, we applied a chronological split. The earliest 80% of rows formed a single training pool (796 samples), and the latest 20% formed one held-out test set (199 samples).

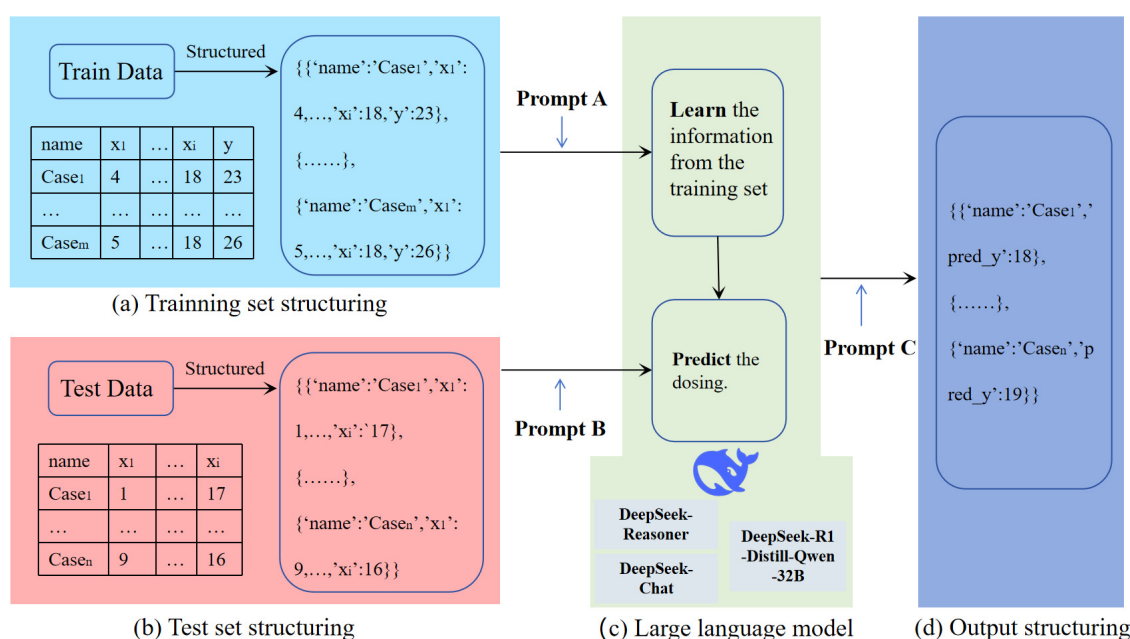


Figure 1. LLM-based framework for coagulant dosage prediction via in-context learning.

Prior to model input, raw data underwent systematic structural reconstruction to align with LLM input requirements and enhance parsing efficiency, as detailed in Figure 1a,b. The “Training-Set Structuring” module converted conventional time-series storage formats (e.g., spreadsheets or relational databases) into a model-readable dictionary structure. Each training sample was restructured into the following format: `{‘name’:‘Case i’,‘x1’:5,...,‘x8’:18,‘y’:26}`, representing the sample ID, multiple flocculation-related attributes, and the target dosage, respectively. The test set underwent identical processing via the “Test-Set Structuring” module, with the key distinction that the dosage key-value pair was excluded (e.g., `{‘name’:‘Case j’,‘x1’:9,...,‘x8’:16}`).

During the prediction phase, we used a fixed three-part prompting strategy (A-B-C) to standardize inference and improve reproducibility across models. The same template was applied to DeepSeek-Reasoner (DeepSeek-V3.2), DeepSeek-Chat (DeepSeek-V3.2), and DeepSeek-R1-Distill-Qwen-32B (Qwen2.5-32B).

Prompt A (instruction + schema + demonstrations) defined the task and structured format, then appended labeled historical cases from the active training partition. In concise form, it followed:

“You are a water treatment expert. Predict dosing from water quality parameters using historical cases.

Variable map: $x_1 = \langle \text{feature}_1 \rangle, \dots, x_8 = \langle \text{feature}_8 \rangle; y = \text{Dosing}.$

Case i: $x_1 = \dots, x_2 = \dots, \dots, x_8 = \dots \rightarrow y = \dots$

Prompt B (target query) then requested a prediction for the focal sample using the same key-value structure:

“Now predict dosing for:

$x_1 = \dots, x_2 = \dots, \dots, x_8 = \dots$ ”

For test samples, no target dosage label was provided.

Prompt C (output constraint for structured extraction) enforced a machine-readable first line:

“Output format is mandatory: first line must be exactly $\langle \text{pred} \rangle \text{NUMBER} \langle / \text{pred} \rangle$ (NUMBER with up to 2 decimals). Then provide concise reasoning.”

This rule enabled deterministic parsing of numeric predictions from free-form responses and ensured one-to-one alignment between each sample ID and its extracted predicted dosage (e.g., $\{\text{'name': 'Case j'}, \text{'pred_y': 22}\}$).

Following extraction, predictive performance was evaluated on the held-out test set under the chronological split protocol using R^2 and RMSE. Training-set comparisons were not treated as the primary cross-model basis for interpretation because, under in-context learning, prompt-provided labeled examples make LLM “training-set” scores not directly comparable to parameter-fitting training metrics of tabular baselines.

2.3. LLM Selection and Experimental Setup

LLMs have undergone rapid development in recent years. Among them, ChatGPT (e.g., GPT-3.5 Turbo and GPT-4/ChatGPT-4o) and DeepSeek (e.g., DeepSeek-Coder, DeepSeek-R1, and DeepSeek-V3) have drawn significant academic attention due to their distinct characteristics in performance, cost, and applicability [36–40]. Extensive studies have systematically compared these two platforms. Jin et al. [37] pointed out that while ChatGPT excels in user-friendly dialog and broad language tasks, DeepSeek may be more suitable for enterprise environments that require cost control, intensive reasoning, or open customization. Furthermore, Diniz-Freitas and Diz-Dios [41] compared the efficacy of DeepSeek-R1 and ChatGPT-4o in oral lesion diagnosis, finding that DeepSeek-R1 achieved a diagnostic accuracy of 91.6%, slightly surpassing ChatGPT-4o (88.9%) and significantly outperforming the journal readership’s average of 47.8%.

To achieve an optimal balance between experimental efficacy and computational cost, this study selected three distinct large language models developed by DeepSeek: DeepSeek-Reasoner, DeepSeek-Chat, and DeepSeek-R1-Distill-Qwen-32B. DeepSeek-Reasoner enhances reasoning capability through reinforcement learning and exhibits robust performance in reasoning and in-context learning [42]. DeepSeek-Chat serves as a powerful Mixture-of-Experts (MoE) model with 671B parameters. Comprehensive evaluations indicate that it outperforms alternative open-source models and achieves performance parity with leading closed-source models [43]. Additionally, DeepSeek-R1-Distill-Qwen-32B was used as a lightweight variant optimized through knowledge distillation. Evaluation results demonstrate that this distilled model delivers exceptional performance despite its reduced parameter scale [42].

Generation was performed via the OpenAI-compatible chat.completions.create interface for all three LLMs. Experiments were conducted during 11–23 April 2026 (local time). The evaluated models were DeepSeek-Chat (API ID: deepseek-chat), DeepSeek-Reasoner (API ID: deepseek-reasoner), and DeepSeek-R1-Distill-Qwen-32B (API ID: deepseek-ai/DeepSeek-R1-Distill-Qwen-32B). In our run window, DeepSeek-Chat and DeepSeek-Reasoner corresponded to the DeepSeek-V3.2 service generation (officially announced upgrade: 1 December 2025). The distilled model (DeepSeek-R1-

Distill-Qwen-32B) was released in January 2025 and is based on Qwen2.5-32B (released 19 September 2024). For DeepSeek-Chat, decoding parameters were temperature = 0.3, top_p = 0.9, and max_tokens = 8192. For reasoning models, temperature/top_p are unsupported; max_tokens = 30,000 was used for both DeepSeek-Reasoner and DeepSeek-R1-Distill-Qwen-32B. Context-length handling followed a fail-fast policy: no additional client-side truncation was applied, and if overflow occurred, the request would fail at the API level; no context-overflow errors were observed in the reported runs. Each reported LLM evaluation used three fixed integer seeds (42, 62, and 82), and all reported R^2 and RMSE values for the three LLMs are arithmetic means across these three runs. To improve robustness, transient API failures were handled with bounded retries (max_retries = 3, retry_delay = 5 s, request_timeout = 600 s), while non-retryable failures were logged and skipped so batch evaluation could continue.

Finally, to benchmark the effectiveness of LLMs in coagulant dosage prediction, we compared their performance with four commonly used ML models in the field—XGBoost [44], ANN [45], Support Vector Machines (SVM) [8], and k-nearest neighbors (k-NN) [16]. This comparison is intended as a task-level operational benchmark under shared evaluation conditions, rather than a claim that in-context learning and conventional model training are mechanistically equivalent. To mitigate temporal leakage and keep the benchmark aligned with the in-context learning (ICL) evaluation, all tabular models used the same chronological train/test partition (earliest 80% as the training pool; latest 20% as a single held-out test set), avoiding random splits that can transmit autocorrelation across folds. Hyperparameter selection was restricted to the training pool and used five-fold expanding-window time-series cross-validation, with features standardized using statistics computed on each training fold only (no statistics from validation or test segments) and hyperparameters chosen by the highest mean validation R^2 across folds; the held-out test segment was excluded from scaling and tuning. All four models were tuned over pre-specified full Cartesian grids; ANN additionally employed early stopping on validation RMSE within each fold (maximum-epoch cap with patience).

2.4. Performance Evaluation Metrics

Following established practices in the field [35,46], this study employed the Coefficient of Determination (R^2) and the Root Mean Square Error (RMSE) to evaluate the predictive accuracy of LLMs and to compare them with alternative ML models. R^2 is a statistical metric that quantifies the proportion of variance in the dependent variable explained by the model. Values closer to 1 indicate better performance. RMSE measures the square root of the average squared differences between predicted values and actual observations, where lower values signify higher prediction accuracy. The mathematical formulations for R^2 and RMSE are presented in Equations (3) and (4), respectively.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - y_{i,predict})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - y_{i,predict})^2} \quad (4)$$

3. Results and Discussion

3.1. SHAP-Based Feature Importance Analysis

Feature importance analysis is crucial for interpreting model dependencies and understanding how input features relate to prediction accuracy. Shapley additive explanations (SHAP) provide a unified game-theoretic framework for attributing the predictions of a

given supervised model to its input features [47], building on earlier game-theoretic explanations of individual predictions [48]. Conceptually, each feature's SHAP value summarizes marginal changes in the model output associated with including versus withholding that feature across coalitions of the other variables; in applied work, these quantities are obtained with scalable estimators—e.g., TreeSHAP for tree ensembles [47]. In this study, TreeSHAP attributions were computed for the XGBoost gradient-boosting regression baseline, evaluated on the 995 retained monitoring samples used in Figure 2. Through this analysis, we quantified the contribution of each feature to each individual prediction outcome.

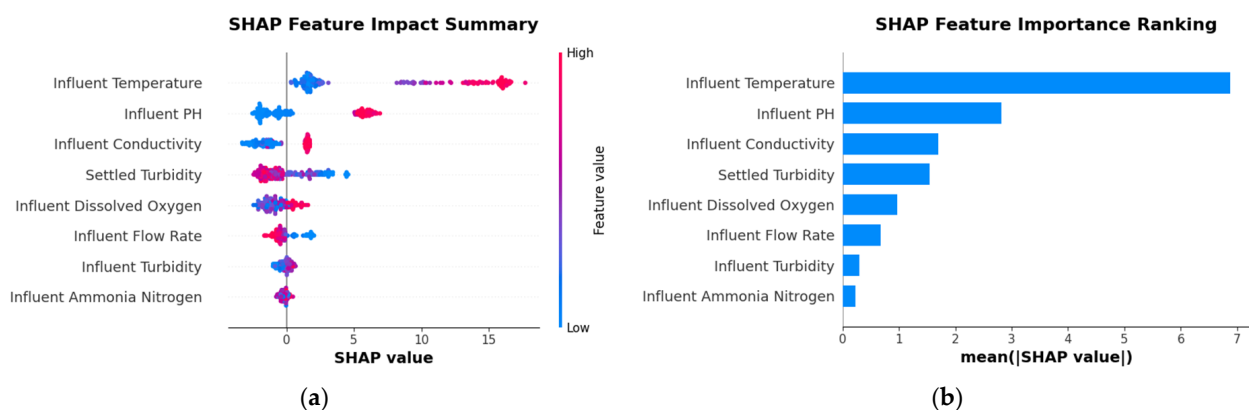


Figure 2. SHAP-based feature importance analysis. (a) SHAP feature impact summary. (b) SHAP feature importance ranking. (TreeSHAP on the fitted XGBoost dosage regressor; $N = 995$ samples).

Figure 2a illustrates how the value of each feature affects the XGBoost-predicted coagulant dosage under TreeSHAP. Each point represents one sample, with color gradients indicating the magnitude of the feature value (red for high values and blue for low values). The position along the x-axis (SHAP value) indicates whether that feature contributes to an increase (positive) or a decrease (negative) in the predicted dosage for that instance.

In this case, the SHAP summary plot shows the following patterns for coagulant dosage. Influent temperature has the strongest positive effect, with high temperatures at large positive SHAP values. Influent pH and influent conductivity are also positively associated with dosage. Settled turbidity is negatively associated: low values tend to raise the prediction, and high values tend to lower it. Influent dissolved oxygen shows a modest positive tendency, with high DO more often on the positive SHAP side. Influent flow rate is negatively associated with high flow, lowering the prediction. Influent turbidity is weakly positive. Influent ammonia nitrogen shows only a weak, near-zero SHAP pattern with little consistent directional effect on dosage.

In this case, Figure 2b ranks features by mean absolute SHAP values: influent temperature is clearly dominant, followed by influent pH. Influent conductivity and settled turbidity form the next tier. Influent dissolved oxygen and influent flow rate are moderately important. Influent turbidity ranks below the leading group in mean $|\text{SHAP}|$ terms, while influent ammonia nitrogen is the least important. Overall, the model is driven mainly by temperature and pH, with the other variables as secondary signals.

3.2. Evaluation of Synergistic Effects of Multi-Parameter Inputs

In this case, SHAP feature importance analysis identified influent temperature as the most influential predictor of coagulant dosage. For consistency with the existing benchmark design and to assess the value of a single feedforward input, this section compares model performance under two settings: (i) the full-feature set and (ii) influent temperature as the sole input.

All R^2 and RMSE entries for DeepSeek-Reasoner, DeepSeek-Chat, and DeepSeek-R1-Distill-Qwen-32B in Table 1 are arithmetic means over three API decoding seeds (42, 62, and 82), as defined in Section 2.3; tabular-model entries are single-run results on the same fixed chronological train/test split, adopted to reduce temporal leakage risk and better reflect operational forecasting conditions. To avoid misleading comparisons, we report and interpret only held-out test-set metrics here, because in in-context learning, LLM “training-set” scores mainly reflect retrieval from prompt-provided labeled examples and are therefore not conceptually comparable to training-fit metrics of tabular models.

Table 1. Results of all features and single features for different models.

Features	Model	Test	
		R^2	RMSE
All features	DeepSeek-Reasoner	0.6213	6.4175
	DeepSeek-Chat	0.6393	6.2617
	DeepSeek-R1-Distill-Qwen-32B	−0.1826	11.3480
	XGBoost	0.6499	6.1736
	ANN	−0.5312	12.9110
	SVM	0.7885	4.7981
	k-NN	0.6421	6.2418
Influent temperature	DeepSeek-Reasoner	0.1913	9.3826
	DeepSeek-Chat	0.1981	9.3432
	DeepSeek-R1-Distill-Qwen-32B	−0.1679	13.3923
	XGBoost	0.3292	8.5454
	ANN	−0.2661	11.7403
	SVM	0.3455	8.4410
	k-NN	0.3158	8.6307

For the all-features setting, selected hyperparameters were: ANN (hidden 32–64, dropout 0.2, lr 0.001, batch size 32, epochs 300, patience 20), SVR (RBF kernel, $C = 50$, epsilon = 0.05, gamma = 0.01), XGBoost (n_estimators = 500, max_depth = 6, learning_rate = 0.1, subsample = 1.0, colsample_bytree = 1.0, random_state = 42), and k-NN ($k = 3$, distance weighting, Minkowski $p = 2$). For the temperature-only setting, selected hyperparameters were: ANN (hidden 32–64, dropout 0.1, lr 0.001, batch size 16, epochs 300, patience 20), SVR (RBF kernel, $C = 10$, epsilon = 0.1, gamma = 0.1), XGBoost (n_estimators = 500, max_depth = 4, learning_rate = 0.1, subsample = 0.8, colsample_bytree = 0.8, random_state = 42), and k-NN ($k = 3$, uniform weighting, Minkowski $p = 1$).

Under this protocol, in the HQ-WTP case dataset (Table 1), with all features, SVM performed best ($R^2 = 0.7885$, RMSE = 4.7981), while XGBoost, k-NN, DeepSeek-Chat, and DeepSeek-Reasoner formed a similar middle tier ($R^2 = 0.6213$ – 0.6499 ; RMSE = 6.1736–6.4175). DeepSeek-R1-Distill-Qwen-32B and ANN were markedly weaker ($R^2 = -0.1826$ and -0.5312 ; RMSE = 11.3480 and 12.9110). Under temperature-only input, performance declined for all models: SVM remained best ($R^2 = 0.3455$, RMSE = 8.4410), XGBoost and k-NN followed closely ($R^2 = 0.3292$ and 0.3158), DeepSeek-Chat and DeepSeek-Reasoner were lower but positive ($R^2 = 0.1981$ and 0.1913), and DeepSeek-R1-Distill-Qwen-32B and ANN remained negative in R^2 . Overall, in this case dataset, the full-feature configuration consistently outperformed the temperature-only configuration, and within the three large language models, DeepSeek-Chat and DeepSeek-Reasoner were similar and clearly stronger than DeepSeek-R1-Distill-Qwen-32B.

3.3. Sensitivity Analysis of Model Performance to Dataset Size

The size of the dataset is a critical factor that can influence the prediction performance of LLMs [35]. Consequently, a sensitivity analysis was conducted based on the primary dataset of 995 samples. This test sample set was kept identical for all training-size conditions (full, 1/2, and 1/4). We then varied only the training data available to each model by using (i) the full training pool, (ii) a one-half (1/2) condition that retains the last 50% contiguous segment of the training pool, or (iii) a one-quarter (1/4) condition that retains the last 25% contiguous segment of the training pool. Tabular models (XGBoost, ANN, SVM, k-NN) were trained and evaluated on the same chronological partitions for comparability. LLM metrics follow the three-seed mean reporting rule in Section 2.3. The experimental results are illustrated in Figure 3.

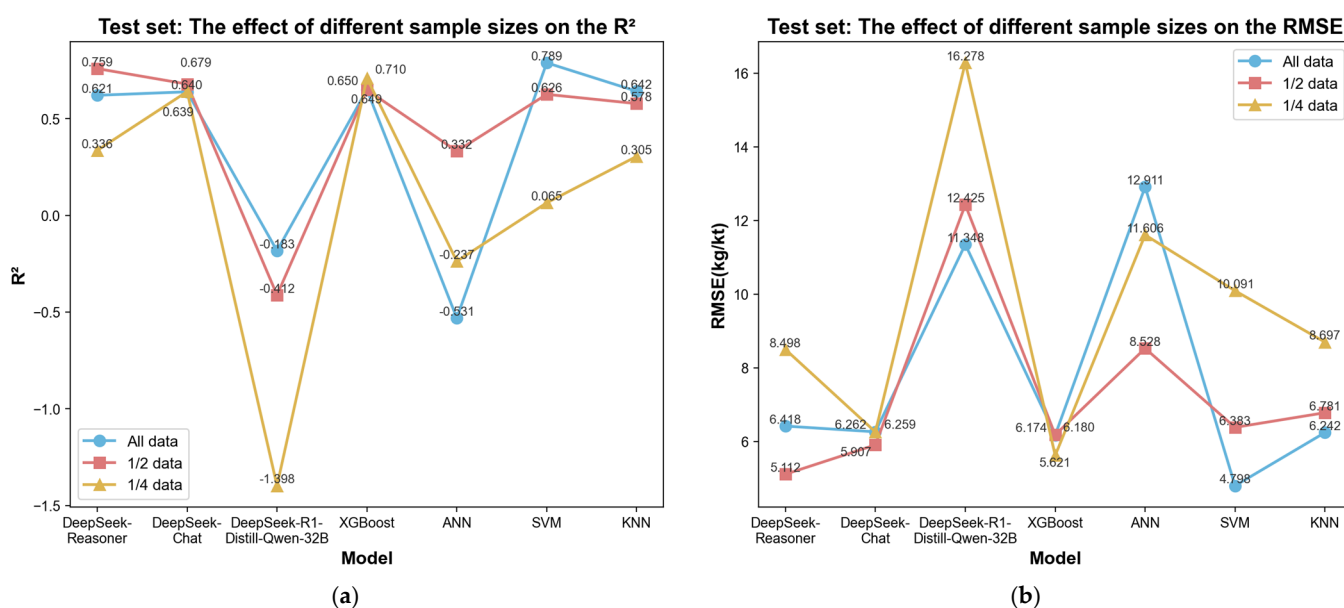


Figure 3. R^2 and RMSE curves of the evaluated models on test sets. (a) R^2 curves on the test set. (b) RMSE curves on the test set.

Figure 3 reports test-set performance only (R^2 and RMSE) under the fixed chronological test partition, with three training-pool conditions (all, 1/2, and 1/4). To avoid misleading comparisons, only held-out test-set metrics are interpreted here.

Across the three LLMs, performance varied by training-pool condition. DeepSeek-Reasoner achieved its best test performance at 1/2 data ($R^2 = 0.759$, RMSE = 5.112), compared with all data ($R^2 = 0.621$, RMSE = 6.418), but dropped at 1/4 data ($R^2 = 0.336$, RMSE = 8.498). DeepSeek-Chat showed relatively stable test behavior, with R^2 around 0.64 for all and 1/4 data (0.639 and 0.640) and a moderate improvement at 1/2 data ($R^2 = 0.679$; RMSE = 5.907). DeepSeek-R1-Distill-Qwen-32B remained the weakest LLM in all three conditions, with negative R^2 throughout (all: -0.183 ; 1/2: -0.412 ; 1/4: -1.398) and sharply increased RMSE at 1/4 data (16.278).

For tabular baselines, model responses to training-pool size were also heterogeneous. XGBoost reached its best test score at 1/4 data ($R^2 = 0.710$, RMSE = 5.621), while SVM performed best with all data ($R^2 = 0.789$, RMSE = 4.798) and degraded as the pool shrank (1/4: $R^2 = 0.065$, RMSE = 10.091). k-NN declined progressively from all to 1/4 data (R^2 : $0.642 \rightarrow 0.578 \rightarrow 0.305$; RMSE: $6.242 \rightarrow 6.781 \rightarrow 8.697$). ANN was unstable across conditions, peaking at 1/2 data ($R^2 = 0.332$, RMSE = 8.528) but remaining negative at all and 1/4 data.

Overall, the test-set results indicate that performance is sensitive to training-pool condition, but the optimal condition is model-dependent rather than universal. For LLMs, these differences should be interpreted as joint effects of training-pool size and in-context prompt composition; therefore, the present benchmark does not support a general claim that smaller or larger training pools are uniformly better across architectures.

3.4. Robustness and Error Analysis of the Models Under Extreme Water-Quality Fluctuations

In practical water-treatment operation, abrupt fluctuations in raw-water quality may appear as atypical samples in the multivariate feature space. To examine model behavior under such non-routine conditions, outliers were identified using Isolation Forest on chronologically split data (earlier period for training and later period for testing), with predictor standardization fitted on the training set only. Under the primary setting (contamination = 0.05), the detected outliers were 40/796 (5.03%) in the training set and 129/199 (64.82%) in the test set.

To assess sensitivity to the outlier definition, we additionally evaluated contamination values of 0.03, 0.05, 0.10, and 0.15. The corresponding training-set outliers were 24/796 (3.02%), 40/796 (5.03%), 80/796 (10.05%), and 120/796 (15.08%), while the test-set outliers were 105/199 (52.76%), 129/199 (64.82%), 140/199 (70.35%), and 154/199 (77.39%), respectively. This monotonic pattern is expected under progressively less conservative contamination settings. At the same time, the relatively high outlier proportion in the later test segment may reflect a temporal distribution shift between earlier and later periods. Therefore, outlier-subset results are treated as supportive robustness evidence and interpreted with caution rather than used alone for strong comparative claims.

As shown in Figure 4, model performance on test-set outliers differed substantially. SVM achieved the best agreement and the lowest error ($R^2 = 0.867$, $RMSE = 3.869$), followed by k-NN ($R^2 = 0.847$, $RMSE = 4.140$). DeepSeek-Chat ($R^2 = 0.774$, $RMSE = 5.039$), DeepSeek-Reasoner ($R^2 = 0.766$, $RMSE = 5.124$), and XGBoost ($R^2 = 0.728$, $RMSE = 5.528$) formed a middle-performance group. DeepSeek-R1-Distill-Qwen-32B showed weaker robustness ($R^2 = 0.257$, $RMSE = 9.142$), and ANN performed worst on this subset ($R^2 = -0.565$, $RMSE = 13.258$). For the three LLMs, the reported values are means over the same three decoding seeds as in Table 1.

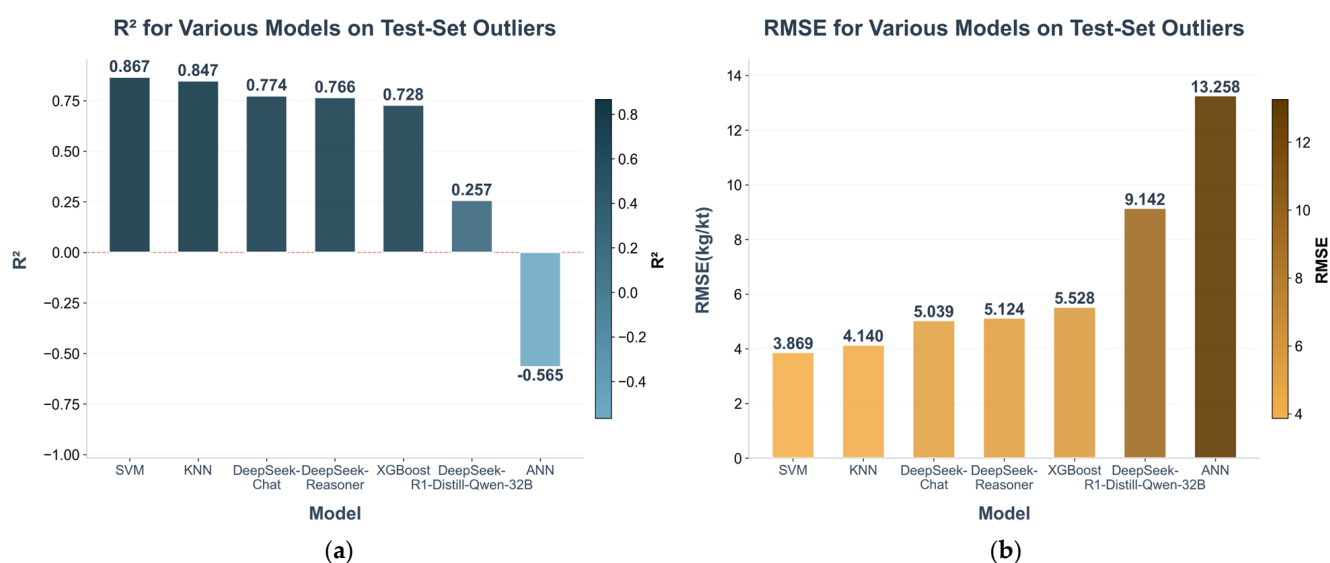


Figure 4. R^2 and RMSE of the evaluated models on test-set outliers. (a) R^2 bar plot on test-set outliers. (b) RMSE bar plot on test-set outliers.

3.5. Interpreting LLM Outputs: Narrative Rationale Patterns and Hallucination Analysis

To characterize how models justify dosage predictions in language, this section examines Chain-of-Thought (CoT) text and accompanying explanations on the test set and summarizes recurring narrative motifs. These codes are derived from surface-generated text only and are not treated as evidence of proprietary internal computation or human-like cognition. Furthermore, we systematically analyzed the underlying origins of prediction deviations, specifically focusing on hallucination as manifested in model outputs.

3.5.1. Analysis of Narrative Rationale Patterns in Model Outputs

To further summarize how models describe dosage choices in natural language, this study categorized verbal patterns in their outputs on the test set, as illustrated in Figure 5. These categories are derived from the analysis and summarization of the models' response content and are treated as descriptive labels of generated text rather than verified internal algorithms. For responses that directly and explicitly describe the method used, we categorize them according to their self-reported approach. For example, when a response states: "Based on historical data, using the k-Nearest Neighbors ($k = 2$) algorithm to predict the dosing amount. . . Selecting the 2 nearest neighbors with the smallest distances: Case 182 (distance 0.85, dosage 33.74) and Case 221 (distance 0.92, dosage 33.55). Taking the average dosage: $(33.74 + 33.55)/2 = 33.645 \approx 33.65$ ", we summarize it as the "k-NN-like narrative strategy (simple-averaging wording)". For responses that do not explicitly state the method but whose content implies a specific logic, we infer and summarize the core operation based on their description. For example, when a response describes: "Based on the provided historical data cases, the water quality parameters of the new sample are most similar to those of Case 218. Its dosage is 34.10 kg/kt, therefore the predicted dosage is 34.10 kg/kt." Its core operation is finding the single most similar sample, which we infer as the "direct matching strategy". The results revealed that models with different parameter scales and training paradigms exhibited distinct distributions of narrative strategy labels in their outputs.

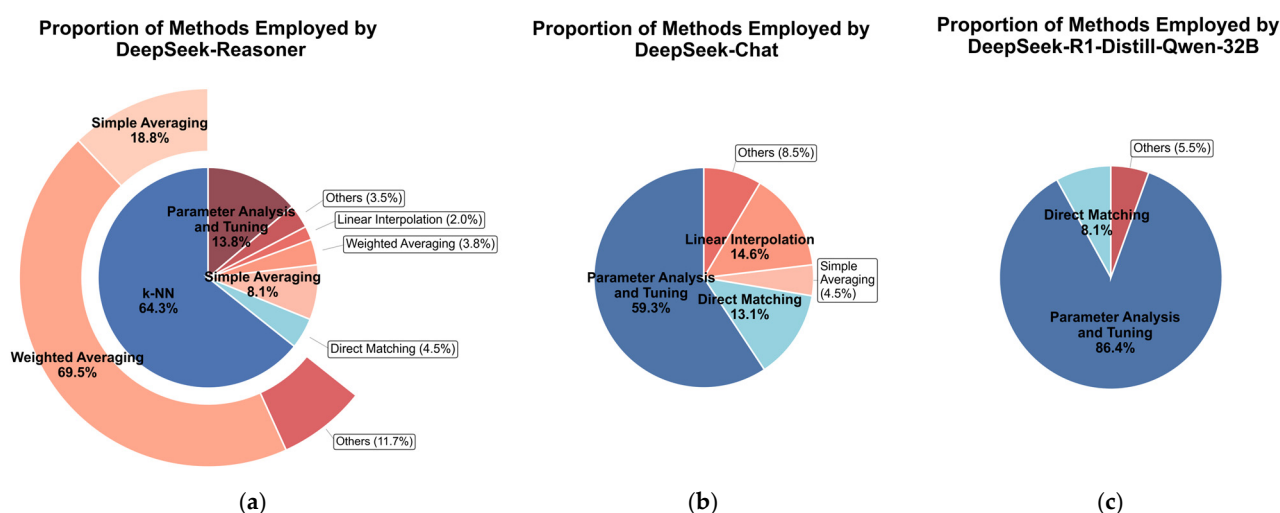


Figure 5. Proportion of narrative strategy labels in model responses for: (a) DeepSeek-Reasoner, (b) DeepSeek-Chat, (c) DeepSeek-R1-Distill-Qwen-32B. In panel (a), the inner ring shows coarse categories, and the outer ring shows sub-label composition; because these are different levels, outer-ring percentages are not additive with inner-ring percentages. Panels (b,c) are single-layer summaries (percentages sum to 100% within each panel).

DeepSeek-Reasoner: At a coarse level, its narrative labels are dominated by k-NN-like patterns, with a prominent weighted-aggregation profile. k-NN-type labels account

for 64.3% overall, including weighted averaging (44.7%), simple averaging (12.1%), and other k-NN-like variants (7.5%). Additional labels include parameter analysis and tuning (13.8%), direct matching (4.5%), standalone simple averaging (8.1%), standalone weighted averaging (3.8%), others (3.5%), and linear interpolation (2.0%). These categories describe output narratives only and do not establish internal computational mechanisms.

DeepSeek-Chat: Based on the coding results, DeepSeek-Chat is primarily characterized by parameter analysis and tuning narratives. Parameter analysis and tuning is dominant (59.3%), followed by linear interpolation (14.6%), direct matching (13.1%), and others (8.5%); simple averaging (4.5%) is relatively infrequent. In this dataset, DeepSeek-Chat's response text is more often framed as feature/parameter reasoning than nearest-case lookup.

DeepSeek-R1-Distill-Qwen-32B: The distilled model is dominated by parameter analysis and tuning narratives (86.4%), with direct matching (8.1%) and others (5.6%) as secondary labels. Simple averaging, weighted averaging, and linear interpolation are 0% in the current coding results. These categories are interpreted as output-text patterns rather than verified internal algorithms.

Overall, the updated coding indicates clear cross-model differences in response-level rationale narratives: parameter-analysis language is dominant in DeepSeek-Chat and DeepSeek-R1-Distill-Qwen-32B, while DeepSeek-Reasoner remains more k-NN-like at the coarse level. These findings are descriptive of generated text and are not interpreted as direct evidence of latent decision mechanisms.

3.5.2. Quantitative Characterization of Hallucination Types and Error Attribution

To more thoroughly assess the reliability of LLMs in coagulant dosage prediction, we systematically categorized hallucination phenomena in the test set and quantified the contribution of each hallucination type to the overall prediction error, as summarized in Figure 6. The definition and statistics of hallucination types in this study are based on operationally defined reasoning inconsistencies or factual contradictions present in the model's output (i.e., the "response" field). Accordingly, this audit evaluates the consistency of generated explanations rather than latent model computation itself. Although coding follows predefined criteria, some residual interpretation subjectivity may remain. The specific classification criteria are as follows:

Calculation Hallucination: Refers to numerical operation errors in the model's stated reasoning text, such as standardization, mean calculation, or rounding errors.

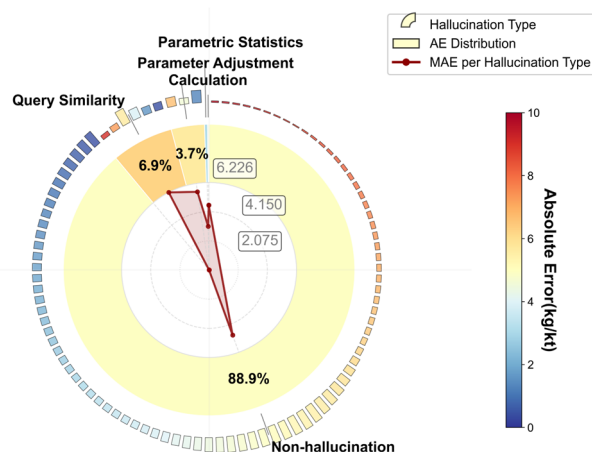
Parametric Statistics Hallucination: Occurs when the model incorrectly reports or references statistical parameters from historical data, such as misstating key statistical information like dosage ranges or water quality parameter distributions.

Parameter Adjustment Hallucination: Refers to instances where the model modifies its predictions based on self-introduced subjective assumptions or unverified adjustment logic, rather than relying on patterns from historical data.

Query Similarity Hallucination: Occurs when the model claims in its response to have referenced a specific historical case, but the mentioned case ID does not exist, or the parameter comparisons it describes do not match the actual situation.

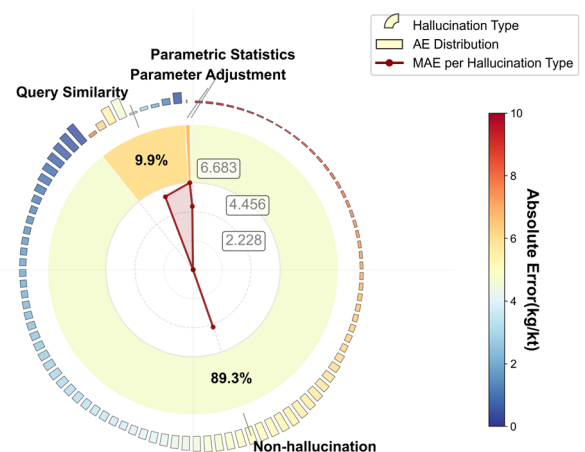
We manually labeled each test-set response using the criteria above, with access to the same model inputs and the historical records or case list needed to verify factual statements in the text. Each case received one primary hallucination label from a primary coder. A second coder independently relabeled a randomly selected subset of test responses ($n = 60$, 30.15% of the 199 test cases) for each model–seed run; we report percent agreement, and Cohen's kappa, and we reconciled disagreements by discussion to reach consensus. Full test-set statistics use the primary coder's final labels; because judgments rely on free-text interpretation, some residual subjectivity remains.

Hallucination Type Distribution and Error Analysis of DeepSeek-Reasoner



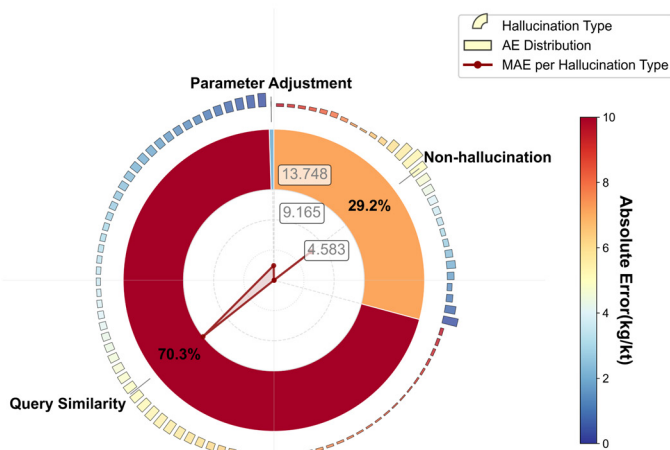
(a)

Hallucination Type Distribution and Error Analysis of DeepSeek-Chat



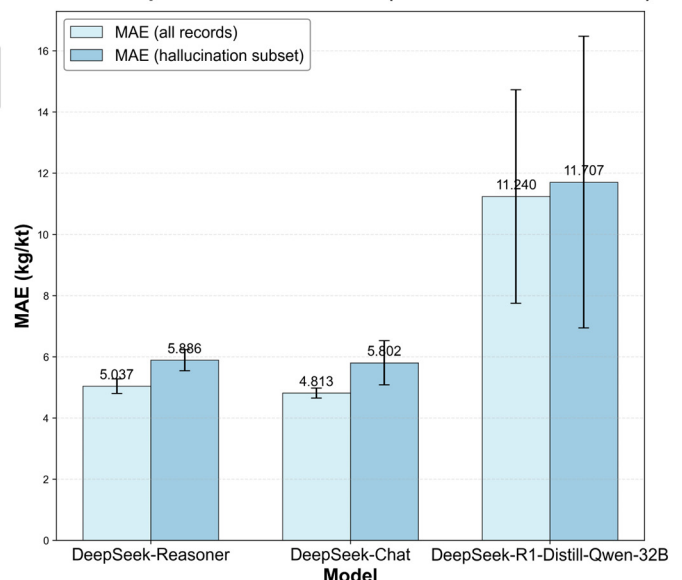
(b)

Hallucination Type Distribution and Error Analysis of DeepSeek-R1-Distill-Qwen-32B



(c)

MAE Comparison Across Models (Mean $\pm$ SD Across Seeds)



(d)

Figure 6. Hallucination type distribution and error analysis of: (a) DeepSeek-Reasoner, (b) DeepSeek-Chat, (c) DeepSeek-R1-Distill-Qwen-32B, and (d) hallucination count and mean absolute error for three models.

Using this taxonomy and coding protocol, we then conducted a comprehensive statistical analysis of hallucination occurrences and their corresponding prediction errors for each model. The quantitative results reveal distinct reliability profiles and failure modes among the models, as detailed below for model-generated rationale text.

DeepSeek-Reasoner shows a relatively low hallucination proportion (11.06%), mainly concentrated in Query Similarity (6.87%) and Parameter Adjustment (3.69%). Query Similarity is also the dominant error-bearing type (type-level MAE approx. 6.23 kg/kt), indicating that case-matching inconsistencies remain the primary risk source in this model. The outer ring in Figure 6a further indicates that most errors are distributed in low-to-mid bins, with a limited high-error tail compared with the distilled model.

DeepSeek-Chat has a similar overall hallucination proportion (10.72%), but with a more concentrated structure dominated by Query Similarity (9.88%). This dominant type

also carries substantial error magnitude (type-level MAE approx. 5.99 kg/kt), suggesting that risk is less about multi-type hallucination spread and more about one key failure mode. In Figure 6b, the outer-ring distribution is still centered in mid-error bins, but its high-error tail is somewhat heavier than that of DeepSeek-Reasoner.

DeepSeek-R1-Distill-Qwen-32B exhibits a markedly elevated hallucination proportion (70.82%), overwhelmingly driven by Query Similarity (70.32%). This dominant type has a very high error magnitude (type-level MAE approx. 13.75 kg/kt), showing that hallucination frequency and severity are strongly coupled in the distilled model. Consistently, Figure 6c shows a clearly heavier high-error tail in the outer-ring distribution than in the other two models, indicating systemic reliability degradation rather than isolated failures.

At the aggregate level (Figure 6d), DeepSeek-Reasoner and DeepSeek-Chat remain close in hallucination proportion, while the distilled model is substantially higher. Cross-seed error statistics further separate robustness levels:

- DeepSeek-Reasoner: MAE(all records) = 5.04 ± 0.24 , MAE(hallucination subset) = 5.89 ± 0.34 kg/kt
- DeepSeek-Chat: MAE(all records) = 4.81 ± 0.17 , MAE(hallucination subset) = 5.80 ± 0.72 kg/kt
- DeepSeek-R1-Distill-Qwen-32B: MAE(all records) = 11.24 ± 3.49 , MAE(hallucination subset) = 11.71 ± 4.77 kg/kt

Together, these results indicate that DeepSeek-Reasoner provides the best balance between hallucination control and stability, DeepSeek-Chat maintains strong baseline accuracy, but with greater hallucination-event variability, the distilled model combines high hallucination exposure with poor robustness.

In conclusion, DeepSeek-Reasoner achieved the best robustness and the lowest hallucination risk in tandem with output narratives that often resemble ensemble analogy-based justification. DeepSeek-Chat produces deeper CoT-style prose but exhibits domain-specific failure blind spots in those narratives. Meanwhile, the distilled small-scale model oversimplifies the narrative pattern, resulting in a marked degradation in reliability. These evaluations provide useful guidance for the selection and optimization of future LLMs in the water sector.

4. Summary and Conclusions

This study assessed the applicability of large language models (LLMs) for coagulant-dosage prediction in drinking-water treatment under a shared chronological benchmark with conventional tabular ML models. The conclusions are based on the HQ-WTP case dataset and should be interpreted within this scope. The proposed LLM workflow is positioned as a task-level decision-support benchmark against XGBoost, ANN, SVM, and k-NN, not as evidence that in-context learning and conventional parameter-fitting are mechanistically equivalent. Therefore, we do not claim replacement of established plant control systems or universal deployment readiness.

Using the full-feature set consistently outperformed using influent temperature alone, indicating that dosage prediction in this case benefits from multivariate process information under nonlinear operating conditions. In the full-feature test setting, SVM was the strongest-performing tabular baseline, while DeepSeek-Reasoner and DeepSeek-Chat showed intermediate performance, and DeepSeek-R1-Distill-Qwen-32B showed lower and less stable performance. LLM test metrics were reported as arithmetic means over three fixed decoding seeds (42, 62, 82).

Performance varied across training-pool conditions (all, 1/2, 1/4), and the optimal condition was model-dependent rather than universal. In this benchmark, DeepSeek-Reasoner achieved its best test performance at the 1/2 condition. Because training-pool size co-varies with prompt length and demonstration composition in in-context learning,

this pattern is interpreted as a protocol-specific joint effect rather than causal evidence that smaller or larger pools are inherently superior.

CoT-style outputs were analyzed as output-level narrative rationale patterns rather than verified internal computation traces. Hallucination risk differed across models, suggesting that lightweight deployment may require stricter reliability controls in safety-relevant contexts.

In practice, the LLM workflow should be used strictly as a decision-support layer with human-in-the-loop review, predefined safety bounds, and automatic fallback when outputs are uncertain or inconsistent. For external API use, data minimization, de-identification, encrypted transmission, role-based access control, and auditable logging are required; private-cloud or on-premise deployment is preferred when confidentiality requirements are strict. Future work should prioritize independent multi-site validation, stronger temporal modeling, standardized uncertainty reporting, and robust hallucination-detection/correction pipelines.

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